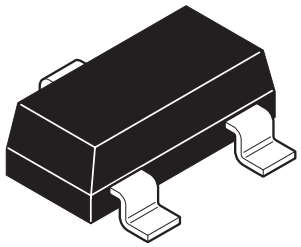
2N7002
60V SOT23 N-channel enhancement mode MOSFET

Summary



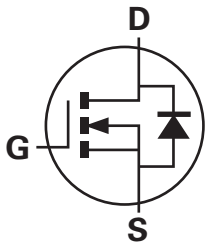
$V_{(BR)DSS}$	$R_{DS(on)}$ (Ω)	I_D (A)
60	7.5 @ V_{GS}= 10V	0.5
	7.5 @ V_{GS}= 5V	0.05

Description

A small signal MOSFET for general purpose switching applications.

Features

- Fast switching speed
- Low gate drive capability
- SOT23 package

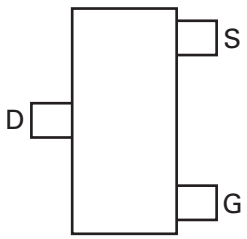


Applications

- General switching applications

Ordering information

Device	Reel size (inches)	Tape width (mm)	Quantity per reel
2N7002	7	8	3,000



Top view

Device marking

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Absolute maximum ratings

Parameter	Symbol	Limit	Unit
Drain-source voltage	V_{DS}	60	V
Continuous drain current at T_{amb}=25°C	I_D	115	mA
Pulsed drain current	I_{DM}	800	mA
Gate-source voltage	V_{GS}	±40	V
Power dissipation at T_{amb} =25°C	P_{tot}	330	mW
Operating and storage temperature range	T_j, T_{stg}	-55 to +150	°C

Electrical characteristics (at T_{amb} = 25°C unless otherwise stated)

Parameter	Symbol	Min.	Max.	Unit	Conditions
Drain-source breakdown voltage	BV_{DSS}	60		V	I_D= 10μA, V_{GS}=0V
Gate-source threshold voltage	$V_{GS(th)}$	1	2.5	V	I_D= 250μA, V_{DS}=V_{GS}
Gate-body leakage	I_{GSS}		10	nA	V_{GS}=±20V, V_{DS}=0V
Zero gate voltage drain current	I_{DSS}		1	μA	V_{DS}= 48V, V_{GS}=0V
			500	μA	V_{DS}= 48V, V_{GS}=0V, T=125°C
On-state drain current[a]	$I_{D(on)}$	500		mA	V_{DS}= 25V, V_{GS}= 10V
Static drain-source on-state voltage[a]	$V_{DS(on)}$		3.75	V	V_{GS}= 10V, I_D= 500mA
			375	mV	V_{GS}= 5V, I_D= 50mA
Static drain-source on-state resistance[a]	$R_{DS(on)}$		7.5	Ω	V_{GS}= 10V, I_D= 500mA
			7.5	Ω	V_{GS}= 5V, I_D= 50mA
Forward transconductance[a][b]	g_{fs}	80		mS	V_{DS}= 25V, I_D= 500mA
Input capacitance[b]	C_{iss}		50	pF	V_{DS}= 25V, V_{GS}=0V f=1MHz
Common source output capacitance[b]	C_{oss}		25	pF	
Reverse transfer capacitance[b]	C_{rss}		5	pF	
Turn-on time[b][c]	$t_{(on)}$		20	ns	V_{DD}≈30V, I_D= 200mA, R_g=25Ω, R_L=150Ω
Turn-off time[b][c]	$t_{(off)}$		20	ns	

NOTES:
(a) Measured under pulsed conditions. Pulse width ≤300μs; duty cycle ≤2%.
(b) Sample test.
(c) Switching times measured with 50Ω source impedance and <5ns rise time on a pulse generator
Spice parameter data is available upon request for this device.

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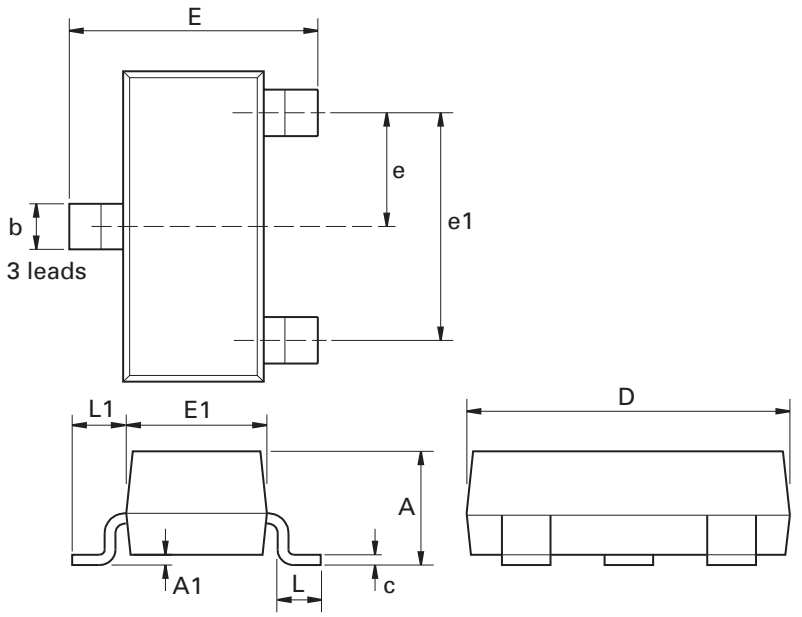
SOT23 Package outline



Dim.	Millimeters		Inches		Dim.	Millimeters		Inches	
	Min.	Max.	Min.	Max.		Min.	Max.	Min.	Max.
A	-	1.12	-	0.044	e1	1.90 NOM		0.075 NOM	
A1	0.01	0.10	0.0004	0.004	E	2.10	2.64	0.083	0.104
b	0.30	0.50	0.012	0.020	E1	1.20	1.40	0.047	0.055
c	0.085	0.20	0.003	0.008	L	0.25	0.60	0.0098	0.0236
D	2.80	3.04	0.110	0.120	L1	0.45	0.62	0.018	0.024
e	0.95 NOM		0.037 NOM		-	-	-	-	-

Note: Controlling dimensions are in millimeters. Approximate dimensions are provided in inches

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